

Network Configuration for Remote Office

CT330

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Version 1.0

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1 Purpose

This document will serve as a guide for configuring a remote office network that is connected to a headquarters site. The topics covered will include network components, components configurations and connectivity testing.

2 Terms and Definitions

LAN- Local Area Network

WAN- Wide Area Network

HQS- Headquarters Site

OFS- Office Site

rt- Router

sw- Switch

VLAN- Virtual Local Area Network

Client Device- Desktop PC

3 Situation and Minimum Requirements

3.1 Situation

A remote office site (OFS) needs to be connected via a WAN connection to the headquarters site (HQS). The OFS site must provide connections for at least 20 client devices.

3.2 Minimum Requirements

The following are the minimum requirements for our network:

The remote office site, OFS, must connect to the Headquarters site HQS

The HQS will have one large switch and one large router

A WAN connection is used to connect the large HQS router to the small OFS router

The OFS will have a small router and 2 medium sized switches

The 2 medium switches at OFS will be connected using an etherchannel trunk

Each LAN segment at the OFS will support a minimum of 20 client devices

Client devices at the HQS and OFS will receive IP addresses via DHCP

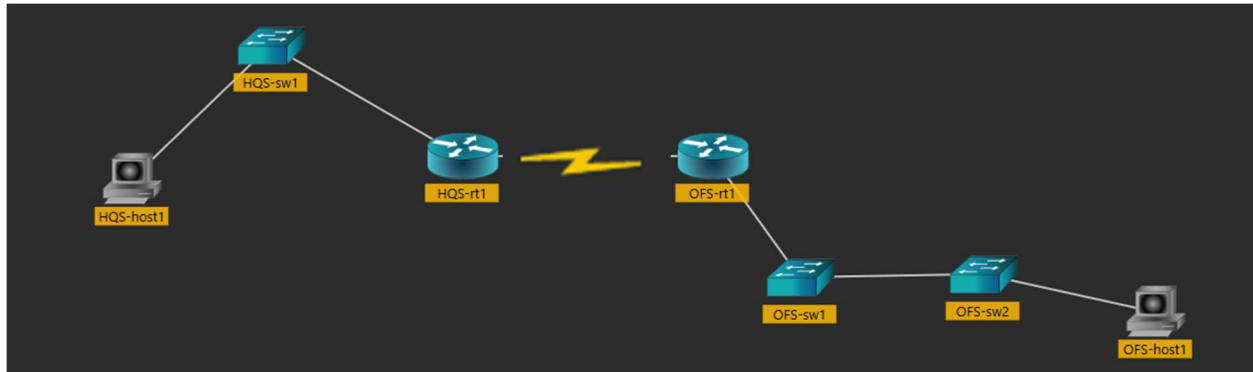
4 Solution

For the HQS a model 2960-48TT-L switch (HQS-sw1) with 48 ports available will be used.

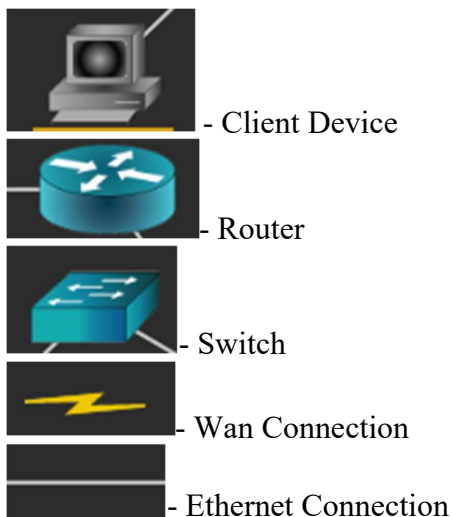
Attached to the HQS-sw1 via Fast Ethernet port will be a model 3640 router (HQS-rt1) with 4 serial ports and 1 Fast Ethernet port. HQS-rt1 will be connected via serial connection to a model 3620 router (OFS-rt1) with 4 serial ports and 4 ethernet ports at the OFS. Attached to OFS-rt1 via Ethernet port will be a model 2960-24TT-L (OFS-sw1) switch with 24 Fast Ethernet ports.

OFS-sw1 will connect via Fast Ethernet to a model 2960-24TT-L (OFS-sw2) switch with 24 Fast Ethernet ports. This will provide connectivity for 24 users on each LAN segment at the OFS. The connection between OFS-sw1 and OFS-sw2 will be a configured as a trunk.

5 Diagram



KEY:



6 Configuration

6.1 Specifications

Commands for HQS-rt1

- enable
- configure terminal
- hostname HQS-rt1
- interface fastethernet1/0 (LAN connection)
- ip address 10.100.1.1 255.255.255.0
- no shutdown
- exit
- interface serial0/0 (WAN connection)
- ip address 10.50.1.1 255.255.255.252

- clock rate 64000
- no shutdown
- exit
- service dhcp
- ip dhcp excluded-address 10.100.1.1
- ip dhcp pool HQS-Clients
- network 10.100.1.0 255.255.255.0
- default-router 10.100.1.1
- exit
- ip route 10.200.1.0 255.255.255.0 10.50.1.2
- end

Commands for HQS-sw1

- enable
- configure terminal
- hostname HQS-sw1
- vlan 1
- name HQS-LAN
- exit
- interface fastethernet0/1
- switchport mode access
- switchport access vlan 10
- no shutdown
- exit
- end

Commands for OFS-rt1

- enable
- configure terminal
- interface serial0/0 (WAN connection)
- ip address 10.50.1.2 255.255.255.252
- no shutdown
- exit
- interface ethernet1/0 (LAN connection)
- ip address 10.200.1.1 255.255.255.0
- no shutdown
- exit
- service dhcp
- ip dhcp excluded-address 10.200.1.1
- ip dhcp pool OFS-Clients
- network 10.200.1.0 255.255.255.0
- default-router 10.200.1.1
- exit
- ip route 10.100.1.0 255.255.255.0 10.50.1.1
- end

Commands for OFS-sw1

- enable
- configure terminal
- vlan 1
- name OFS-LAN
- interface fastethernet0/1
- switchport mode access
- switchport mode vlan 1
- no shutdown
- exit
- interface range fastethernet0/2 – 3
- channel-group 1 mode active
- exit
- interface Port-channel1
- switchport mode trunk
- no shutdown
- exit
- end

Commands for OFS-sw2

- enable
- configure t
- vlan 1
- name OFS-LAN
- exit
- interface range fastethernet0/2 – 3
- channel-group 1 mode active
- exit
- interface Port-channel1
- switchport mode trunk
- no shutdown
- exit
- end

6.2 Verification

Testing DHCP configurations

- Attach one Client device to the HQS-sw1.
- Attach one Client device to the OFS-sw2.
- On each client device, enter the following commands to attain an IP address via DHCP and to view the current IP address.
- C:> ipconfig /ip dhcp
- C:> ipconfig

```

C:>ipconfig /ip dhcp
C:>ipconfig

HELP
    Manipulates ip information for Workstation.

    IPCONFIG [/all | /dg | /dns | /ip | /ipv6 | /release | /renew]
    /all      Display full configuration information
    /dg       Adds the default gateway to the workstation
    /dns      Adds the dns server to the workstation
    /ip       Adds the ip address and subnet mask to the workstation
    /release  Release the IPv4 address to the workstation
    /renew    Renew the IPv4 address to the workstation
    /release6 Release the IPv6 address to the workstation
    /renew6   Renew the IPv6 address to the workstation

Examples:
    ? ipconfig /ip 157.1.1.12 255.0.0.0
    ? ipconfig /dg 157.1.1.1

Boson BOSS 5.0 IP Configuration
Ethernet adapter Local Area Connection:
    Connection-specific DNS Suffix . : 
    IP Address. . . . . : 10.200.1.11
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.200.1.1
C:>

```

- This confirms that DHCP is properly configured to assign IP addresses when a Client device connects to the network.

Testing connection from OFS client to HQS router

- On the OFS client run the following commands
- C:> ping 10.100.1.1
- C:> tracert 10.100.1.1

```

C:>ping 10.100.1.1

Pinging 10.100.1.1 with 100 bytes of data:
Reply from 10.100.1.1: bytes=100 time=18ms TTL=15
Reply from 10.100.1.1: bytes=100 time=3ms TTL=15
Reply from 10.100.1.1: bytes=100 time=2ms TTL=15
Reply from 10.100.1.1: bytes=100 time=2ms TTL=15
Reply from 10.100.1.1: bytes=100 time=2ms TTL=15

Ping statistics for
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 18ms, Average = 5ms
C:>tracert 10.100.1.1

Tracing route to 10.100.1.1.
   1          9 ms          3 ms          2 ms    10.200.1.1
   2          7 ms         13 ms          6 ms    10.50.1.1

Trace complete.

```

- This confirms that an OFS client can communicate with the HQS router.

Testing connection from HQS client to OFS router

- On the HQS client run the following commands
- C:> ping 10.200.1.1
- C:> tracert 10.200.1.1

```
C:>ping 10.200.1.1

Pinging 10.200.1.1 with 100 bytes of data:
Reply from 10.200.1.1: bytes=100 time=3ms TTL=15
Reply from 10.200.1.1: bytes=100 time=4ms TTL=15
Reply from 10.200.1.1: bytes=100 time=4ms TTL=15
Reply from 10.200.1.1: bytes=100 time=1ms TTL=15
Reply from 10.200.1.1: bytes=100 time=2ms TTL=15

Ping statistics for
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 3ms
C:>tracert 10.200.1.1

Tracing route to 10.200.1.1.
   0  1 ms    1 ms    1 ms  10.100.1.1
   1  2 ms    1 ms    1 ms  10.50.1.2
   2  1 ms    2 ms    4 ms  10.200.1.1

Trace complete.
```

- This confirms that an HQS client can communicate with the OFS router.

7 References

BOSON Lab- Router Basics
BOSON Lab- IP Addressing
BOSON Lab- Configuring Router Interfaces
BOSON Lab- Switch Basics
BOSON Lab- Configuring VLANs
BOSON Lab- Configuring Trunking
BOSON Lab- Testing Connectivity